Extraction and Analysis of Caffeine and Aflatoxins in Food Matrices Using HPLC and Observational Study of GC-MS

An Industrial Placement Report

Submitted By

Name: Aszad Hossain Akib

Roll: 20/28

Reg No: 03396

Session: 2019-2020

A Report Submitted for Partial Fulfillment for the Degree of Bachelor of Science in Food Science and Technology

Chattogram Veterinary and Animal Sciences University

Chittagong-4225

February 2025

Extraction and Analysis of Caffeine and Aflatoxins in Food Matrices Using HPLC and Observational Study of GC-MS

An Industrial Placement Report

Submitted By

Name: Aszad Hossain Akib

Roll: 20/28

Reg No: 03396

Session: 2019-2020

Approved as to style and contents by

.....

(Signature of the Author)

Aszad Hossain Akib

.....

(Signature of the Co Supervisor)

Mohammad Tariqul Hasan

Principle Scientific Officer

Food Toxicology Research Section

Institute of Food Science and Technology

BCSIR

.....

(Signature of the Supervisor)

Prof. Dr. Indrajit Saha

Department of Physical and Mathematical Sciences

Faculty of Food Science and Technology

CVASU

Report Organization

S.L. No.	TITLE	PAGE NO.
1.	Acknowledgement	I.
2.	Abstract	II.
3.	Table of Contents	III.
4.	List of Tables	IV.
5.	List of Figures	V.
6.	List of Abbreviation	VI.

Acknowledgement

I am grateful to ALLAH for giving me strength, wisdom, and perseverance to complete my own placement at a desired institution.

I thank my supervisor, Professor **Dr. Indrajit Saha**, Department of Physical and Mathematical Sciences at Chattogram Veterinary and Animal Sciences University (CVASU), for his invaluable guidance and unwavering support, which shaped my internship journey. His dedication and encouragement were motivating during challenging times, and I am grateful for the skills I gained under his mentorship.

I am indebted to Professor **Dr. Ferdusee Akter**, Dean of the Faculty of Food Science and Technology, CVASU, for her feedback, encouragement, and support. Her expert advice has enriched my work, and I appreciate her time and effort in reviewing my study.

I would like to express my sincere gratitude to **Ms. Ayesha Begum**, Associate professor in the Department of Applied Food Science and Nutrition, Faculty of Food Science and Technology, Chattogram Veterinary and Animal Sciences University, for her invaluable guidance, support, and encouragement throughout my internship.

I thank the scientific officers at the Food Toxicology Research Section, IFST, BCSIR, including **Sheik Muzahidul Islam, Maksuda Begum, and Puja Saha**. Their support, technical assistance, and insights were crucial in completing my project, and I value your expertise during the experimental phases and data analysis.

A special mention to my BCSIR supervisor, **Mohammad Tariqul Hasan**, for his guidance and support throughout the research. His knowledge and recommendations enhanced the quality of my work, and I am grateful for his mentorship.

I acknowledge the faculty, staff, and administrators of Chattogram Veterinary and Animal Sciences University and BCSIR for providing the facilities and resources for my research.

Finally, I thank my family for their love, support, and encouragement. Their belief in me and understanding during difficult times have been my foundation. I am blessed to have such a wonderful family who inspires me to achieve my goals.

The author

February, 2025

Abstract

This report focuses on analyzing aflatoxins and caffeine in food matrices, specifically spices, chanachur, cereal-based products, coffee, and soft drinks. Aflatoxins are toxic secondary metabolites produced by *Aspergillus* species and are potent carcinogens commonly found in food products like cereals and spices. In this study, aflatoxins were extracted from food samples using the QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe) method, which is a widely used technique for pesticide residue extraction, and quantified using High-Performance Liquid Chromatography with Fluorescence Detection (HPLC-FLD). Caffeine, a naturally occurring stimulant found in beverages such as coffee and soft drinks, was extracted using sonication and analyzed using HPLC with Photodiode Array Detection (HPLC-PDA). The results showed significant variation in caffeine levels between coffee (29119 ppm) and soft drinks (104.04 ppm), with coffee containing a much higher concentration of caffeine. This study underscores the importance of HPLC in accurately quantifying both aflatoxins and caffeine, ensuring food safety, and adhering to regulatory standards. Furthermore, challenges such as optimizing extraction methods for complex matrices like high-fat foods and spices were addressed. The study demonstrates that HPLC with fluorescence detection is reliable for detecting aflatoxins and caffeine in diverse food matrices. The findings contribute to the development of analytical techniques to ensure food safety.

Keywords: Aflatoxins, Caffeine, QuEChERS, High-Performance Liquid Chromatography (HPLC), Fluorescence Detection (FLD), Photodiode Array Detection (PDA), Food Safety, Extraction methods, Spices, Analytical techniques

Table of Contents

CHAPTER One 1
Introduction 1
CHAPTER Two 3
Aims and Objectives 3
CHAPTER Three 4
Literature Review: 4
CHAPTER Four 7
Industrial Functionality 7
4.1 Background of the Industrial Placement 7
4.1.1 BCSIR (Bangladesh Council of Scientific and Industrial Research). 7
4.1.2 Introduction to IFST (Institute of Food Science and Technology) 9
4.1.3 Food Toxicology Research Section (FTRS) 11
4.2 Instrumentation and Working Principle of HPLC and GC-MS: 14
4.2.1 HPLC: 14
4.2.2 GC-MS: 20
Chapter Five 24
Materials and Method 24
5.1 Materials: 24
Reagents: 24
Apparatus: 24
5.2 Methods: 24
5.2.1.5 Running on HPLC with FLD Detector 26
5.2.2.4 Running on HPLC with PDA Detector 27
5.4 Identification and Quantification of Caffeine in Coffee by HPLC 29
5.4.1 Calibration curve for Caffeine 29
5.4.2 HPLC curve for Coffee 30
5.5 Identification and Quantification of Caffeine in soft drinks 31
5.5.1 HPLC curve for SOFT DRINKS 32
Chapter 6 34
Result and Discussion 34
Chapter 7 36
Conclusion 36
References 37

List of Tables

List of Tables	Page no.
Table 1: Focus Areas of BCSIR	8
Table 2: Institutes at BCSIR	9
Table 3: Research Divisions of IFST	10
Table 4: Research Section of IFST	11
Table 5: List of Instruments in FTRS	13
Table 6: Appendix	40

List of Figures

List of figure	Page no.
Figure 1: Chemical Structure of Aflatoxin	1
Figure 2: Chemical Structure of Caffeine	2
Figure 3: Working principle of HPLC	17
Figure 4: Analytical column	17
Figure 5: Column Specifications	18
Figure 6**:** Agilent HPLC system with manual injector.	18
Figure 7: Jasco HPLC system with autosampler	19
Figure 8: Chromatogram of Caffeine	20
Figure 9: Calibration Curve of Caffeine	28
Figure 10: Working Principle of GC-MS	22
Figure 11: **:** Shimadzu GC-MS-QP2010 Plus	23
Figure 12: GC-MS Column	23
Figure 13: Structure of Caffeine	29
Figure 14: Calibration Curve of Caffeine	29
Figure 15:Chromatogram of Cafffeine in Coffee	30
Figure 16: Chromatogram of Cafffeine in Soft Drinks	32

List of Abbreviation

List of abbreviation is given below

1. **BCSIR** - Bangladesh Council of Scientific and Industrial Research
2. **IFST** - Institute of Food Science and Technology
3. **FTRS** - Food Toxicology Research Section
4. **HPLC** - High-Performance Liquid Chromatography
5. **FLD** - Fluorescence Detection
6. **PDA** - Photodiode Array Detection
7. **QuEChERS** - Quick, Easy, Cheap, Effective, Rugged, and Safe
8. **dSPE** - Dispersive Solid-Phase Extraction
9. **GC-MS** - Gas Chromatography-Mass Spectrometry
10. **SPE** - Solid-Phase Extraction
11. **PSA** - Primary Secondary Amine
12. **C18** - Octadecylsilane
13. **UHT** - Ultra High Temperature
14. **MS/MS** - Mass Spectrometry/Mass Spectrometry
15. **LC-MS/MS** - Liquid Chromatography-Mass Spectrometry/Mass Spectrometry
16. **UV/Vis** - Ultraviolet/Visible
17. **R^2** - Correlation Coefficient
18. **NaCl** - Sodium Chloride
19. **$MgSO_4$** - Magnesium Sulfate

CHAPTER One

Introduction

Aflatoxins are highly toxic secondary metabolites produced by *Aspergillus flavus* and *Aspergillus parasiticus*, commonly found in agricultural commodities like peanuts, corn, and other cereal-based products, especially when these foods are stored improperly under humid conditions. Aflatoxins are potent carcinogens that pose significant health risks, with long-term exposure linked to liver toxicity, cancer, and other severe health conditions. These toxic metabolites are of particular concern in regions where such food products are consumed frequently [1]. Therefore, detecting and quantifying aflatoxins in food products are essential to safeguard public health. One widely adopted method for aflatoxin extraction from complex matrices like cereals, spices, and processed foods is QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe). This method employs liquid-liquid extraction combined with solid-phase extraction (SPE), effectively cleaning up the sample to ensure that the interferences from the food matrix do not affect the final analysis [2]. Following extraction, the quantification of aflatoxins is commonly carried out using High-Performance Liquid Chromatography (HPLC) with fluorescence detection (FLD), which offers high sensitivity even at trace concentrations, making it ideal for aflatoxin detection in complex food matrices [3].

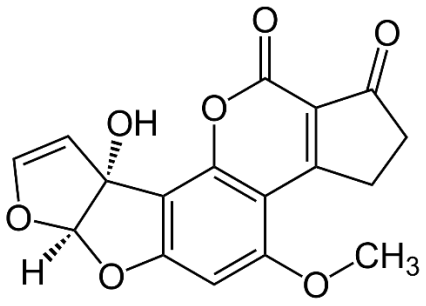


Figure 1: Chemical Structure of Aflatoxin

Caffeine, a naturally occurring stimulant found in plants such as coffee, tea, and cocoa, is consumed globally through beverages like coffee, tea, and soft drinks. While moderate caffeine intake is generally considered safe, excessive consumption can lead to adverse health effects such as increased heart rate, anxiety, and sleep disturbances [4]. The regulation of caffeine content is essential, particularly in energy drinks, where its high

concentration may exceed safe consumption levels. The extraction of caffeine typically involves methods like sonication combined with methanol, followed by purification using solid-phase extraction, before analysis with HPLC equipped with a Photodiode Array Detector (HPLC-PDA). This technique allows for precise quantification of caffeine levels in various beverage matrices, ensuring that caffeine content remains within regulatory limits [5,6].

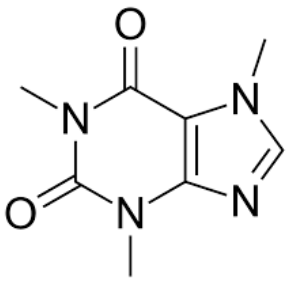


Figure 2: Chemical Structure of Caffeine

The research presented in this report focuses on extracting and analyzing aflatoxins and caffeine from food matrices, including spices, chanachur, cereal-based products, coffee, and soft drinks. Aflatoxins were extracted using the QuEChERS method, and their concentration was determined using HPLC with fluorescence detection (HPLC-FLD). Caffeine was extracted using sonication and methanol, followed by quantification using HPLC-PDA. Additionally, an observational study of Gas Chromatography-Mass Spectrometry (GC-MS) was included, aimed at understanding the principles of this powerful technique for analyzing volatile and semi-volatile compounds in food matrices. This study can help understand the application of established method for extraction and detection of aflatoxin and caffeine in food matrices (spices, chanachur, cereal-ased products, coffee and soft drinks) and hence understanding the instrumentation and working principles of HPLC and GC-MS. This study highlights the importance of reliable, efficient extraction methods and advanced analytical techniques in ensuring the safety of food products.

CHAPTER Two

Aims and Objectives

The aims and objectives of the study are:

1. Applying the QuEChERS method for the extraction of aflatoxins from various food matrices, including spices, chanachur, and cereals.
2. Enhancing the extraction efficiency of caffeine from coffee and soft drink samples, optimizing the process for improved recovery.
3. Observing and establishing the calibration curve for caffeine using HPLC-PDA to ensure accurate quantification.
4. Quantifying aflatoxins using HPLC-FLD and caffeine using HPLC-PDA, employing high-performance liquid chromatography for precise analysis.
5. Gaining a comprehensive understanding of HPLC instrumentation, its working principles, and standard operating procedures (SOPs), while familiarizing with GC-MS instrumentation and its operational principles for potential future applications in contaminant analysis.

CHAPTER Three

Literature Review:

Aflatoxins, produced by *Aspergillus* species, are potent carcinogens commonly found in food products such as spices, chanachur, and cereal-based products. These toxins pose significant health risks, necessitating accurate and efficient extraction and detection methods. The QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe) method has been widely adopted for extracting aflatoxins due to its simplicity and effectiveness, especially in complex matrices. The methods employed in this study, particularly QuEChERS for aflatoxin extraction and HPLC for both aflatoxin and caffeine analysis, have evolved over several decades. The QuEChERS method, introduced in 2003, was initially designed to simplify pesticide residue analysis in fruits and vegetables. Its ability to extract various contaminants, including aflatoxins, has led to its widespread adoption in food safety research [1]. The integration of HPLC with fluorescence and photodiode array detection has further refined the sensitivity and specificity of contaminant analysis, enabling researchers and regulators to detect minute levels of toxins like aflatoxins and caffeine in complex food matrices. Historically, aflatoxin detection required labor-intensive methods involving immunoassays and older chromatographic techniques that lacked the precision needed for modern food safety standards. Similarly, the determination of caffeine in beverages has advanced from basic colorimetric tests to the sophisticated use of HPLC, which provides high-resolution analysis and can simultaneously handle multiple samples. One study developed a modified QuEChERS approach combined with high-performance liquid chromatography-fluorescence detection (HPLC-FLD) to simultaneously determine multiple mycotoxins, including aflatoxins B1, G1, G2, and M1, in milk [7]. The method achieved recovery rates between 73% and 114%, demonstrating its effectiveness in complex matrices, and similar results were observed for extracting aflatoxins from cereal-based products [7]. Further research validated a QuEChERS-based method coupled with ultra-high-performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) for detecting aflatoxins and ochratoxin A in ultra-high-temperature (UHT) and powdered milk [3]. This method showed recoveries ranging from 72% to 121%, suggesting it is suitable for routine analysis of dairy products. A similar approach was applied to spices, which are highly complex matrices with oil content, and showed promising results for aflatoxin extraction [8]. However, matrix effects from high-fat foods, like chanachur, require further optimization of the QuEChERS method to

enhance recovery and minimise interferences. In addition to chromatographic techniques, immunoassays, such as enzyme-linked immunosorbent assays (ELISA), have been explored for aflatoxin detection. Another study reviewed these methods and highlighted their low interference, simple sample pretreatment, and high recovery rates, though issues such as cross-reactivity remain a challenge [6]. This approach provides a non-destructive and fast alternative for aflatoxin detection in various food matrices, including spices. Caffeine is a widely consumed stimulant in beverages such as coffee and soft drinks. Accurate quantification of caffeine is essential for regulatory compliance and quality control. One study introduced an innovative online extraction technique coupled with liquid chromatography-photodiode array detection (LC-PDA) for determining caffeine content in coffee, tea, and cocoa [5]. This method significantly reduced analysis time and solvent usage, streamlining caffeine quantification. Researchers developed a high-performance liquid chromatography method with photodiode array and fluorescence detection (HPLC-PDA-FLD) to simultaneously determine caffeine and water-soluble vitamins in energy drinks [6]. The method demonstrated high sensitivity with limits of detection ranging from 13 to 121 ng/mL, enabling rapid analysis of multiple compounds within a short timeframe. Another study proposed a salting-out assisted liquid-liquid extraction (SALLE) method followed by HPLC with ultraviolet-visible detection (HPLC-UV/Vis) for analysing caffeine and nicotinic acid in coffee [9]. This approach achieved detection limits as low as 0.20 mg/L for caffeine, offering a reliable and efficient alternative for routine analysis in coffee. Recent advancements aim to reduce solvent usage and processing time. This ensures accuracy and reliability in coffee and soft drink matrices for caffeine quantification. The researchers developed a method involving a 60:40 methanol/acetonitrile (%, v/v) extraction mixture, achieving recoveries in the range of 71.8–107.7% at spike levels of 1.5–50.0 µg/kg with relative standard deviations (RSDs) lower than 15% [10]. One study presented a simple, fast, and selective method based on a modified QuEChERS approach for the simultaneous determination of trigonelline, caffeine, and chlorogenic acid in green coffee beans. The method achieved recoveries of 101% for caffeine, demonstrating its potential for routine analysis in the coffee industry [11].

Despite significant advancements in extraction and detection methodologies, challenges remain in optimising protocols for diverse food matrices, including spices, chanachur, and cereal-based products. Applying QuEChERS to high-fat and spice-rich foods requires further refinement to improve aflatoxin recovery rates and minimise matrix effects.

Standardising sonication-assisted extraction methods for caffeine in various beverage matrices requires comprehensive research. This study addresses these gaps by refining extraction techniques and enhancing detection accuracy, contributing valuable insights to food safety and quality control practices.

CHAPTER Four

Industrial Functionality

4.1 Background of the Industrial Placement

This industrial placement was conducted at the Institute of Food Science and Technology (IFST), which is part of the Bangladesh Council of Scientific and Industrial Research (BCSIR), within the Food Toxicology Research Section (FTRS). The primary objective of the placement was to gain hands-on experience in detecting and analyzing toxic substances in food and feed products. During the placement, the focus was placed on the analysis of aflatoxins and caffeine in food samples, using well-established analytical methods. Aflatoxins, which are potent carcinogens produced by Aspergillus species, are commonly found in food products such as cereals, spices, and snacks, and can pose significant health risks, including liver cancer. Therefore, the detection and control of aflatoxin levels in food products are critical for ensuring food safety. Caffeine, a naturally occurring stimulant found in coffee, tea, and soft drinks, was the second focus of this research, as it is important to monitor caffeine levels to ensure they remain within safe consumption limits. The work conducted in this placement included using methods like QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe), dispersive solid-phase extraction (dSPE), and high-performance liquid chromatography (HPLC) for the extraction and analysis of these contaminants. This placement provided a comprehensive learning experience, enhancing knowledge in food safety, public health, and the importance of regulatory compliance in ensuring that food products are safe for consumption. The experience also contributed to strengthening my understanding of food toxicology and the role of advanced analytical techniques in detecting harmful substances in food.

4.1.1 BCSIR (Bangladesh Council of Scientific and Industrial Research).

BCSIR (Bangladesh Council of Scientific and Industrial Research) is an autonomous organization under the Ministry of Science and Technology, Government of Bangladesh. It is the largest and most diverse science and industrial research organization in the country. BCSIR operates eight institutes and three multidisciplinary regional laboratories. These institutes focus on various fields, including R&D programs to address public and private sector challenges and solve fundamental and applied problems.

Focus Areas:

The research activities of BCSIR cover a wide range of areas, including:

Area	Description
Arsenic Mitigation	Research on the removal and mitigation of arsenic contamination.
Analytical Research	Conducts research on various analytical methods.
Animal Food Technology	Research in food safety and quality control in animal food products.
Biogas Technology	Focus on biogas production and its use as an energy source.
Biotechnology	Research in biotechnological advancements.
Ceramic and Chemical Metrology	Development and research in ceramics and chemical measurements.
Conservation of Energy	Focus on energy conservation and exploring new energy sources.
Electronics	Research and development in the field of electronics.
Environmental Pollution	Study of environmental pollution and development of mitigation strategies.
Fats and Waxes; Leather Research & Development	Research on fats, waxes, and leather, and their development.
Food Microbiology and Food Science	Study of microorganisms in food and food safety practices.
Nutrition and Quality Control	Research in nutrition and quality control in food products.
Industrial Products; Pilot Plant Technology	Development of industrial products and pilot plant technologies.

Table 1: Focus Areas of BCSIR

Institutes

BCSIR comprises the following eight institutes:

Institute	Description

Institute of Fuel Research & Development (IFRD)	Focuses on fuel technology research and development.
Institute of Food Science & Technology (IFST)	Specializes in food science, food safety, and nutrition research.
Institute of Glass and Ceramic Research & Testing (IGGCRT)	Research in glass and ceramic materials, testing, and development.
Leather Research Institute (LRI)	Specializes in leather technology and innovation.
Institute of Mining, Mineralogy and Metallurgy (IMMM)	Research on mining, mineralogy, and metallurgy.
Pilot Plant and Process Development Centre (PP&PDC)	Focus on the development of pilot-scale production processes.
Reference Institute for Chemical Measurements	Provides accurate chemical measurements for various scientific applications.
Institute of Nuclear Science and Technology (INST)	Specializes in nuclear science and related technologies.

Table 2: Institutes at BCSIR

4.1.2 Introduction to IFST (Institute of Food Science and Technology)

The **Institute of Food Science and Technology (IFST)** is the largest R&D organization in the country, focused on food science and technology. It transfers scientific technologies to commercial sectors and offers technical services, especially to the seafood industry. The institute is equipped with advanced laboratories and has 17 research sections, managed efficiently through its administrative structure.

Vision

To become a center of excellence in Food Science and Technology by 2041.

Mission

To ensure safe and nutritious food for the people of Bangladesh and support the food industry by promoting scientific technologies.

Research Divisions

Following are the research divisions:

Category	Description
Technology of Plant Food Products	Focuses on the development and improvement of plant-based food products.
Technology of Animal Food Products	Focuses on the development of products derived from animals and their processing technologies.
Technology of Food Grains, Legumes, and Oilseeds	Specializes in research related to grains, legumes, and oilseeds and their food processing.
Microbiology	Conducts research on the role of microorganisms in food safety, quality, and preservation.
Biochemistry and Applied Nutrition	Focuses on the biochemistry of food components and their impact on human health and nutrition.
Food Science and Quality Control	Ensures that food products meet safety, quality, and nutritional standards.
Industrial Development and Rural Technology	Researches innovations that promote industrial development and rural technological advancements.

Table 3: Research Divisions of IFST

Research Sections

Following are the research sections:

Category	Description
Vegetable Technology Research Section	Focuses on vegetable product development and processing technologies.
Fruit Technology Research Section	Specializes in fruit product innovation, preservation, and processing techniques.
Plant Protein Research Section	Researches plant-based proteins, including their extraction, applications, and nutritional value
Category	Description.
Milk Dairy and Food Products Research Section	Specializes in dairy and other food product development, focusing on quality control and nutrition.
Meat, Poultry, and Slaughter House Waste Research Section	Investigates methods for meat processing, poultry, and handling of slaughterhouse waste.

Cereal Technology Research Section	Conducts research on cereal grains and their conversion into food products.
Baby Food, Supplementary Food, and Legume Products Research Section	Focuses on developing products for infants and supplementary food systems.
Oil Seed and Lipid Technology Research Section	Specializes in research related to oilseed processing and lipid-based food products.
Food Microbiology Research Section	Researches the role of microorganisms in food safety, fermentation, and spoilage.
Industrial Microbiology Research Section	Focuses on microorganisms used in industrial food processes, including fermentation and waste management.
Applied Nutrition Research Section	Investigates the nutritional value of foods and how they impact health and wellbeing.
Food Enzymology Research Section	Studies enzymes involved in food processing and their impact on food structure and nutrition.
Quality Control Research Section	Ensures that food products adhere to safety and quality standards throughout production.
Food Toxicology Research Section	Investigates the presence of harmful substances, like aflatoxins, in food products and their impact on health.
Pilot Plant Research Section	Focuses on scaling up food processing techniques for commercial applications.
Fish Technology Research Section	Focuses on seafood product development and technologies for preserving and processing fish products

Table 4: Research Section of IFST

4.1.3 Food Toxicology Research Section (FTRS)

The Food Toxicology Research Section at the Institute of Food Science and Technology (IFST) under the Bangladesh Council of Scientific and Industrial Research (BCSIR) is dedicated to ensuring food safety through

various research initiatives. Their projects focus on detecting and analyzing harmful substances in food products. Key research areas include:

- Mycotoxins: Studying toxic compounds produced by fungi in food.
- Antibiotics: Detecting antibiotic residues in food, particularly in seafood such as shrimp and fish.
- Pesticides: Measuring pesticide residues in food to ensure they stay within safe limits.
- Heavy Metals: Analyzing food for contamination by heavy metals, which can pose serious health risks.

List of Instruments in FTRS:

Instrument	Description
LC-MS/MS	Liquid Chromatography-Mass Spectrometry (LC-MS/MS) is used for detecting and quantifying contaminants in food samples, providing high sensitivity and precision.
GC	Gas Chromatography (GC) is used for separating volatile and semi-volatile compounds for subsequent analysis.
GC-MS	Gas Chromatography-Mass Spectrometry (GC-MS) combines the separation power of GC with the detection capability of MS to analyze complex food matrices.
AAS	Atomic Absorption Spectroscopy (AAS) is used for detecting and quantifying metals in food samples.
HPLC	High-Performance Liquid Chromatography (HPLC) is used for the separation and quantification of various food contaminants, including mycotoxins and caffeine.
Instrument	Description
Refrigerated Centrifuge	Used for sample separation through centrifugation at low temperatures to preserve sample integrity.
-80-degree Refrigerator	Ensures the preservation of sensitive samples at ultra-low temperatures.

Fume Hood	Used for handling toxic substances safely, ensuring proper ventilation during hazardous processes.
Ultra Torex Homogenizer	Used for the homogenization of food samples, ensuring even sample consistency before analysis.
Rotary Evaporator	Used to remove solvents under reduced pressure, a key step in concentration and purification of food samples.
Orbital Shaker	Used to agitate food samples and chemicals for thorough mixing during extraction processes.
Reciprocal Mixture	Used to achieve thorough mixing and extraction of compounds from food samples.
pH Meter	Measures the acidity or alkalinity of food samples during preparation for analysis.
Digital Balance	Provides precise measurement of sample mass for accurate experimental results.

Table 5: List of Instruments in FTRS

4.2 Instrumentation and Working Principle of HPLC and GC-MS:

4.2.1 HPLC:

HPLC is an advanced analytical technique used to separate, identify, and quantify components in samples. Introduced in the 1970s, it enhances traditional column chromatography with high pressure for faster, more precise separations. In toxicology labs like BCSIR under IFST, HPLC is vital for detecting trace contaminants, drugs, and toxins in samples such as blood, urine, and food.

4.2.1.1 Working Principle of HPLC

High-Performance Liquid Chromatography (HPLC) is an advanced chromatographic technique that is used to separate, identify, and quantify components in a mixture. HPLC is based on the principle of differential interaction between the sample components and a stationary phase, facilitated by the movement of a liquid mobile phase under high pressure. This technique provides a high level of resolution and precision, making it widely used for analyzing complex mixtures in fields such as pharmaceuticals, environmental testing, food safety, and biochemical research. The working principle can further be described in the following sections:

a. Mobile Phase and Stationary Phase Interaction

HPLC operates by passing a liquid sample through a column that is packed with a stationary phase (usually silica or polymer-based materials). The stationary phase remains fixed in the column while the mobile phase, typically a liquid, flows through it under pressure. The sample is introduced into the system and carried through the column by the mobile phase.

The components of the sample interact differently with the stationary phase. These interactions can be based on various properties, such as:

- **Polarity**: More polar compounds tend to interact more strongly with polar stationary phases and therefore take longer to elute.
- **Size**: Larger molecules may be separated based on their size if a size-exclusion column is used.

- **Charge**: Ionized compounds may interact with charged stationary phases in ion-exchange chromatography.

As the sample components move through the column, their differing affinities for the stationary phase cause them to travel at different speeds. This differential migration results in the separation of the components, which are then collected and analyzed.

b. High-Pressure Operation

One of HPLC's defining characteristics is the use of high pressure, typically around 3000 psi (pounds per square inch). Pressure is essential for the efficient movement of the mobile phase through the column, which is necessary for the separation of analytes. The high pressure allows for smaller particles in the stationary phase, resulting in more efficient and faster separations.

Smaller particles create more surface area for interactions between the stationary phase and the sample components, which increases resolution and decreases the time required for separation. Applyingw high pressure enables HPLC to handle complex mixtures and produce sharp, well-resolved peaks in the chromatogram.

c. Separation Process

Once the sample enters the column, its components are separated as they interact differently with the mobile and stationary phases. The components that have a stronger interaction with the stationary phase will move more slowly through the column, while those that interact less will elute (exit) faster. This differential interaction results in the separation of components along the length of the column. The rate at which the components are eluted depends on several factors, including the flow rate of the mobile phase, the size and type of the stationary phase, the solvent composition, and the temperature.

The time it takes for a compound to move through the column is known as its retention time. Each compound in the mixture will have a unique retention time, which can be used to identify the compounds when compared to known standards.

d. Detection and Analysis

As the separated components exit the column, they are typically detected using various detection methods, such as:

- **UV/Vis Absorbance**: Most commonly used for organic compounds that absorb ultraviolet or visible light.
- **Fluorescence Detection**: Used for compounds that can fluoresce under UV light.
- **Refractive Index Detection**: Measures changes in the refractive index of the eluent, useful for compounds that do not absorb UV light.
- **Mass Spectrometry (MS)**: Provides highly sensitive detection and structural analysis by identifying the mass-to-charge ratio of compounds.

The detected signals are recorded and displayed as peaks in a chromatogram, where each peak corresponds to a component in the mixture. The area under each peak is proportional to the concentration of the component, allowing for quantitative analysis.

4.2.1.2 Types of HPLC

There are various types of HPLC depending on the type of stationary phase and the interactions involved:

- **Reverse-Phase HPLC (RP-HPLC)**: The most commonly used type of HPLC, where the stationary phase is nonpolar and the mobile phase is polar. The analytes are separated based on their hydrophobic interactions.
- **Normal-Phase HPLC (NP-HPLC)**: The stationary phase is polar, and the mobile phase is nonpolar. This type is typically used for separating nonpolar compounds.
- **Ion-Exchange HPLC**: The stationary phase is charged, and it separates analytes based on their ionic properties.
- **Size-Exclusion HPLC (SEC)**: Separates molecules based on their size, often used for large molecules such as proteins.

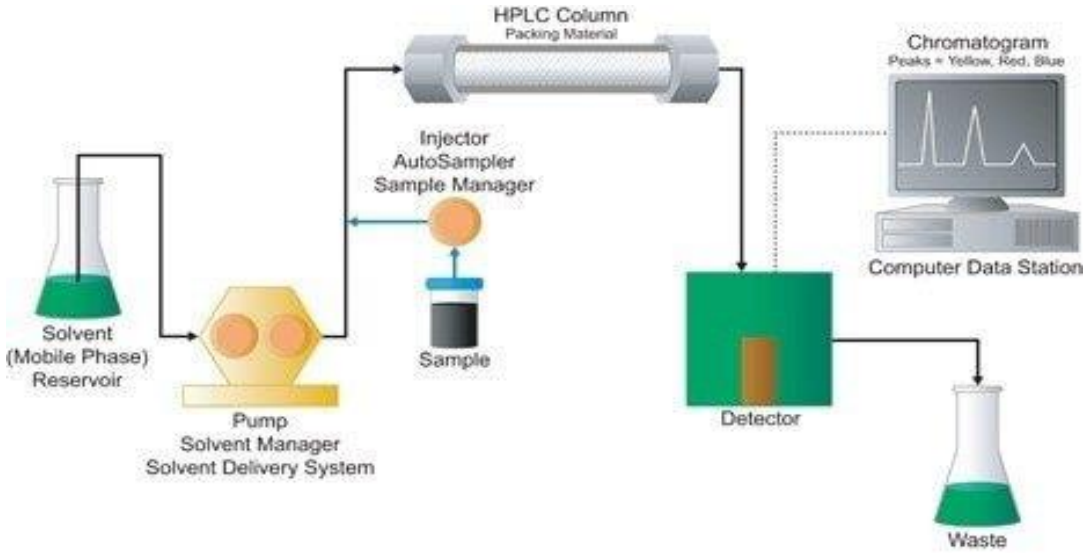


Figure 3: Working principle of HPLC

4.2.1.3 Instrumentation of HPLC

1) Solvent Reservoir: Stores the mobile phase, used in isocratic or gradient elution modes.

2) Pump: Maintains consistent flow and pressure for precise separations.

3) Injector: Introduces samples, either manually (Agilent 1100 with UV/Fluorescence **detectors) or automatically (systems with PDA and Fluorescence detectors).**

4) Detectors:

- **UV/Visible Detector**: Not used in this study.
- **Fluorescence Detector**: Detects aflatoxins by fluorescence under UV light.
- **PDA (Photodiode Array) Detector**: Used for caffeine quantification and spectral analysis.

5) Chromatographic Column: Contains the stationary phase (silica or alumina) for separating sample components. Includes analytical columns for separation and guard columns for protection.



Figure 4: Analytical column

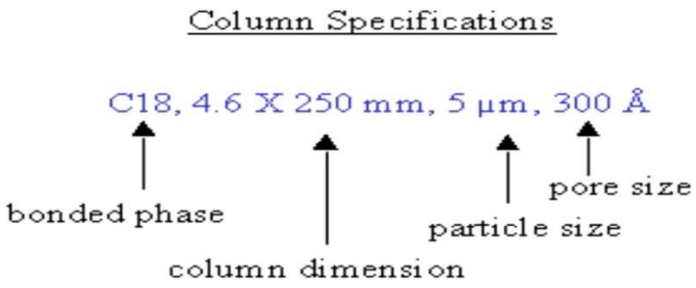


Figure 5: Column Specifications

6) **Data Acquisition System**: Collects and analyzes data using software like ChromPass which is essential tool in analyzing and determining compounds.



Figure 6: Agilent HPLC system with manual injector.



Figure 7: Jasco HPLC system with autosampler

4.2.1.4 Identification and Quantification Mechanisms in HPLC

In HPLC, components are separated based on their different interactions with the stationary and mobile phases. Hydrophobic (nonpolar) compounds move faster, while hydrophilic (polar) compounds are retained longer, leading to their separation as they elute from the column.

a) Using Retention Time for Identification

Retention time is a key parameter in HPLC and refers to the time it takes for a component to travel through the column from the point of injection to the detector. It is used to identify compounds based on their specific retention characteristics.

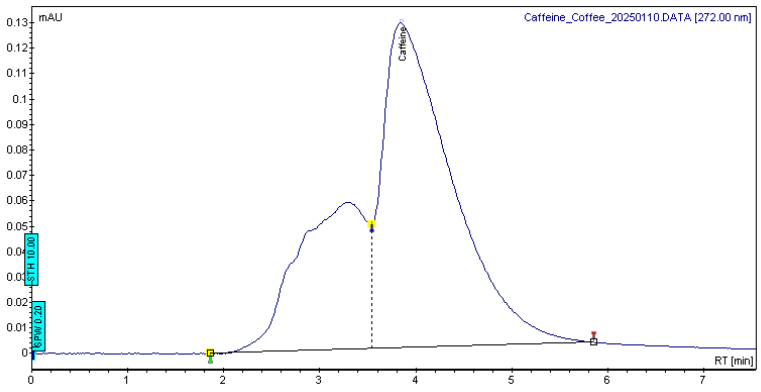


Figure 8: Chromatogram of Caffeine

b) Using Calibration Curve for Quantification

In HPLC quantification, the concentration of an analyte is determined by measuring the area under the chromatographic peak. A calibration curve is created using known concentrations with the equation:

$$y = mx + c$$

Where y is the peak area, m is the slope (representing sensitivity), x is the analyte concentration, and c is the intercept (ideally zero). By measuring the peak area (y) of the sample, its concentration (x) can be calculated by rearranging the equation as:

$$x = (y - c)/m$$

This method ensures accurate quantification based on the calibration curve.

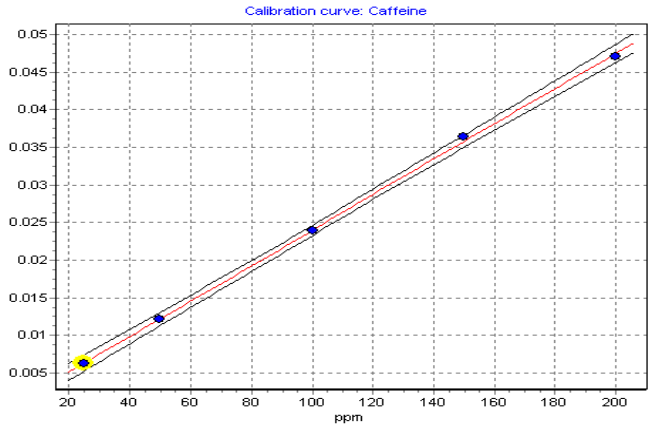


Figure 9: Calibration curve of Caffeine

4.2.2 GC-MS:

4.2.2.1 Principle of GC-MS

Gas Chromatography-Mass Spectrometry (GC-MS) is an analytical technique that combines the separation capabilities of Gas Chromatography (GC) with the detection power of Mass Spectrometry (MS). This system is widely used to analyse volatile and semi-volatile compounds in complex mixtures, offering high sensitivity and precision. The process involves two key steps: separation of components by GC and identification/quantification by MS.

In GC, compounds are separated based on their polarity using a nonpolar stationary phase, typically silica gel coated with polyethene glycol. The mobile phase, an inert gas (usually helium), carries the sample through the column. The separation of compounds is achieved by manipulating the temperature of the column during the run, which allows for varying retention times and optimal separation of analytes.

Once the sample components are separated in the GC column, they are directed into the Mass Spectrometer (MS). The MS operates under vacuum conditions and ionizes the compounds via electron impact (70 eV), generating ions from the neutral molecules. These ions are then analyzed based on their mass-to-charge (m/z) ratio. The resulting spectra are

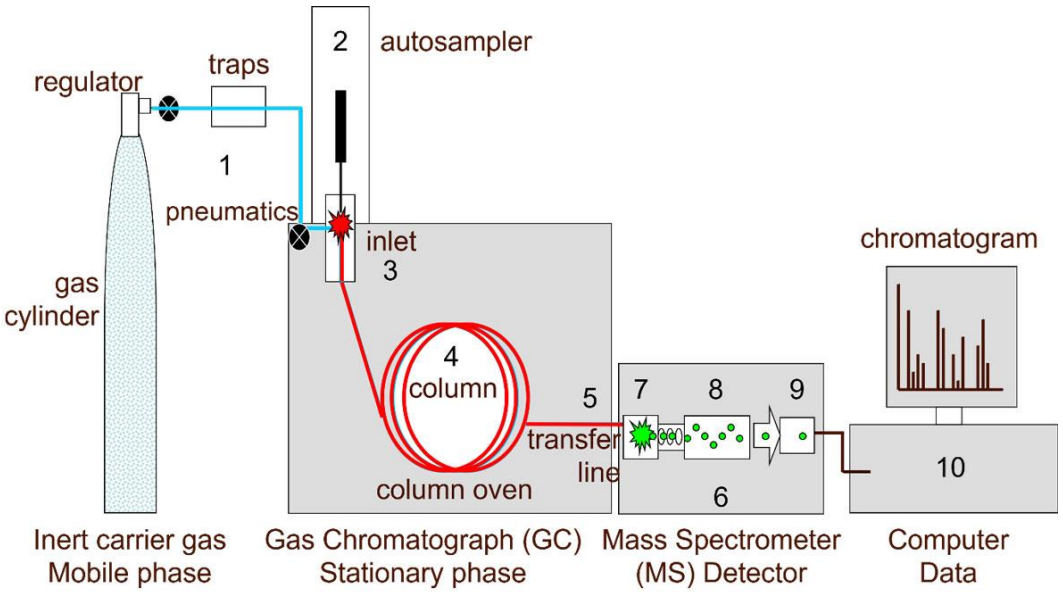


Figure 10: Working Principle of GC-MS

compared against a GC-MS library, which helps identify the compounds present in the sample.

GC-MS can be operated in two modes:

- **Scan Mode**: This mode provides a broad analysis, scanning across a wide range of m/z ratios to detect all ions within the sample.
- **Selected Ion Monitoring (SIM) Mode**: This mode focuses on specific ions of interest, offering greater sensitivity and allowing for targeted analysis of known compounds.

The GC-MS system thus provides both qualitative (identification) and quantitative (measurement of compound concentrations) data, making it invaluable in analytical applications in fields such as toxicology, environmental monitoring, and forensic science.

4.2.2.2 Instrumentation of GC-MS

The Shimadzu GC-MS-QP2010 Plus is the complete system that integrates Gas Chromatography (GC) and Mass Spectrometry (MS) to separate, identify, and quantify compounds in complex mixtures. This system is composed of several key components that work together to perform these tasks:

a) Gas Chromatograph (GC)

The key components of the GC include:

- **Sample Injector**: The sample, typically in liquid or solid form, is vaporized before being introduced into the column via the injector, which is heated to ensure efficient vaporization.
- **Chromatographic Column (Stationary Phase)**: The capillary column is coated with a stationary phase, often silica or polyethylene glycol, that separates compounds based on their differing affinities for the stationary phase and their boiling points. The separation is achieved by allowing the compounds to move through the column with the carrier gas.
- **Carrier Gas**: An inert gas, typically helium, is used as the mobile phase to carry the sample through the column. The flow rate of the carrier gas is carefully controlled for precise separation.
- **Temperature Programming**: The temperature of the column is programmed to optimize the separation of compounds. The temperature can be adjusted during the run to improve retention times for different compounds, aiding in their separation.

b) Mass Spectrometer (MS)

After the compounds are separated in the GC, they are introduced into the mass spectrometer for ionization and analysis. The components of the MS include:

- **Ion Source**: The separated compounds enter the ion source where they are ionized, typically by electron impact (70 eV). This process creates charged ions from the sample molecules.
- **Mass Analyzer**: The mass analyzer sorts the ions based on their mass-to-charge ratio (m/z). In the Shimadzu GC-MS-QP2010 Plus, the mass analyzer is a quadrupole type, which filters ions according to their m/z values and sends them to the detector.
- **Detector**: The detector measures the intensity of the ions and generates a mass spectrum. The mass spectrum helps in identifying the components by comparing the m/z values of the detected ions with known data from libraries.

c) Data Acquisition System

The data acquisition system, including software like LabSolutions**,** collects the data from the detector. The system allows the user to process and analyze the data to identify compounds and quantify their concentrations. The software enables qualitative analysis

(identifying compounds based on their mass spectrum) and quantitative analysis (measuring the abundance of specific ions to determine compound concentrations).

d) Vacuum System

The mass spectrometer operates under a high vacuum to ensure efficient ion movement. The vacuum system reduces the pressure within the MS to a level where ions can travel freely without interference from air molecules. This helps maintain the integrity of the ionization and analysis process.



Figure 11: Shimadzu GC-MS-QP2010 Plus



Figure 12: GC-MS Column

Chapter Five

Materials and Method

5.1 Materials:

Reagents:

- HPLC Grade Solvents: Deionized water, Methanol, Acetonitrile
- Magnesium Sulfate ($MgSO_4$)
- Sodium Chloride (NaCl)
- Primary Secondary Amine (PSA)
- C18

Apparatus:

- Measuring Cylinder
- Beaker
- 0.45 µm Nylon Syringe Filter
- 0.22 µm Nylon Syringe Filter
- 4 mL Glass Vial
- HPLC Vial

5.2 Methods:

5.2.1 Extraction of Aflatoxin Aflatoxin

5.2.1.1 Preparation of Mobile Phase for Aflatoxin Analysis

The mobile phase for aflatoxin analysis was prepared using HPLC-grade solvents**,** including deionized (DI) water, methanol, and acetonitrile**.** The solvent ratio was maintained at H_2O: Methanol: Acetonitrile = 60:20:20**.** To prevent cross-contamination, separate measuring cylinders were used for each solvent.

The preparation involved adding deionized water to a glass container, gradually mixing methanol, and finally incorporating acetonitrile to minimize phase separation. The prepared mixture was sonicated for 30 minutes to eliminate dissolved gases. Once sonication was complete, the mobile phase was transferred into an amber-colored bottle, labeled appropriately, and stored for subsequent use.

5.2.1.2 Preparation of Extraction Solvent for Aflatoxin

The extraction solvent for aflatoxin was also formulated using HPLC-grade solvents**,** specifically deionized water, methanol, and acetonitrile**,** with a ratio of Methanol: Acetonitrile: H_2O = 51:09:40**.**

The required quantity of deionized water was first added to a clean glass container to prepare the solvent. Next, methanol was gently mixed into the water, followed by acetonitrile**,** ensuring minimal phase separation. The solution was then sonicated for 30 minutes to remove air bubbles. After sonication, the extraction solvent was transferred into a labeled glass bottle and set aside for further use.

5.2.1.3 Aflatoxin Extraction Procedure

Aflatoxin extraction is based on the Solid-Liquid Extraction (SLE) principle, where the target compound is separated from a solid sample by dissolving it in a suitable solvent.The extraction procedure follows QuChERS method. The efficiency of the extraction process is influenced by the compound's solubility in the chosen solvent, with higher solubility leading to better extraction.

For the extraction process:

1. A 5-gram powdered sample was accurately weighed and placed into a 50 mL falcon tube.
2. 15 mL of aflatoxin extraction solvent (Methanol: Acetonitrile: H_2O = 51:09:40) was added to the falcon tube.
3. The tube was then shaken vigorously and vortexed for 1 minute to ensure proper mixing.
4. To aid in phase separation, 1 g of NaCl and 2 g of $MgSO_4$ were introduced into the falcon tube.
5. The mixture was centrifuged at 4°C and 5000 rpm for 5 minutes.
6. After centrifugation, 2 mL of the supernatant was carefully collected and transferred to a 15 mL falcon tube for further processing.

5.2.1.4 Aflatoxin Cleanup Process

To purify the extracted aflatoxin sample, 0.075 g of PSA (Primary Secondary Amine) and 0.125 g of C18 (Octadecylsilane) were added to the 15 mL falcon tube containing the 2 mL supernatant. The tube was then:

1. Shaken vigorously and vortexed for 1 minute for thorough mixing.
2. Centrifuged at 4°C and 5000 rpm for 5 minutes.
3. After centrifugation, the purified supernatant was collected .
4. The prepared sample was syringe-filtered to remove any particulates.
5. The filtered solution was transferred into an HPLC vial for analysis.

5.2.1.5 Running on HPLC with FLD Detector

The purified aflatoxin extract was analyzed using high-performance liquid chromatography (HPLC), which is equipped with a fluorescence detector (FLD)**.** The FLD detector was used due to its high sensitivity in detecting aflatoxin compounds, ensuring accurate quantification. The mobile phase was pumped through the HPLC system, and the sample was injected to separate and detect aflatoxin peaks.

5.2.2 Extraction of Caffeine.

5.2.2.1 Preparation of Mobile Phase for Caffeine

The mobile phase for caffeine extraction consists of H_2O: Methanol in a 50:50 ratio**.** To prepare it, the required volume of deionized water was first measured and poured into a glass container**.** Then, methanol was carefully added and mixed gently with water to ensure homogeneity. The prepared solution was sonicated for 30 minutes to eliminate air bubbles. After sonication, the mobile phase was transferred to an amber glass bottle**,** labeled, and stored for future use.

5.2.2.2 Extraction from Coffee Sample

1. A 1 g sample of coffee was weighed and dissolved in 100 mL of hot water to create a coffee solution.
2. The solution was then allowed to cool to room temperature.

3. To enhance caffeine extraction, the cooled solution underwent sonication for a duration of 30 minutes.
4. Following sonication, 1 mL of the solution was transferred into a vial.
5. 1 mL of methanol was added to the vial.
6. The vial was vortexed for 1 minute to mix the components thoroughly.

5.2.2.3 Extraction from Soft Drinks Sample

1. 50 mL of the energy drink sample was measured and taken.
2. The sample was subjected to 30 minutes of sonication to enhance caffeine extraction.
3. After sonication, 1 mL of the energy drink was carefully transferred into a vial.
4. 1 mL of methanol was added to the energy drink sample vial.
5. The vial was vortexed for 1 minute to ensure complete mixing.
6. The prepared sample was syringe-filtered to remove any particulates.
7. The filtered solution was transferred into an HPLC vial for analysis.

5.2.2.4 Running on HPLC with PDA Detector

1. The purified aflatoxin extract was analyzed using high-performance liquid chromatography (HPLC), which is equipped with a Photodiode Array detector (PDA)**.** The PDA detector was used due to its high sensitivity in detecting aflatoxin compounds, ensuring accurate quantification. The mobile phase was pumped through the HPLC system, and the sample was injected to separate and detect caffeine peaks

5.4 Role of Salts, PSA, and C18 in QuChERS-based extraction:

In QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe) based extraction, the use of salts, PSA (Primary Secondary Amine), and C18 plays a crucial role in the efficient extraction and cleanup of samples, particularly in the analysis of contaminants such as pesticides, mycotoxins, and other toxic substances from food matrices. Here's a breakdown of their roles:

a)Salts (Magnesium Sulfate - MgSO4, Sodium Chloride - NaCl):

Salting-Out Effect**:** The addition of salts like magnesium sulfate and sodium chloride promotes the phase separation during the liquid-liquid extraction step. Salts help "salt out" the analytes from the sample matrix, enhancing the extraction of target compounds into the organic solvent phase.

Water Removal: Magnesium sulfate, in particular, is hygroscopic, meaning it absorbs water. This helps in removing excess water from the sample, which is essential for improving the efficiency of extraction and ensuring that the organic solvent is not diluted by water, which would decrease the extraction performance.

b)PSA (Primary Secondary Amine):

Cleanup and Purification: PSA is a solid-phase extraction (SPE) material that is used to clean up the sample after extraction. PSA effectively removes interferences such as sugars, organic acids, and other polar compounds that may be present in the food matrix. It is particularly useful for removing matrix components like fatty acids and polar contaminants that could otherwise interfere with the accurate analysis of target compounds (e.g., mycotoxins or pesticides).

Selective Interaction: PSA has a primary and secondary amine group that interacts with certain matrix components (especially polar ones), allowing the target analytes to remain in the extract while undesirable components are removed. This selective cleanup ensures that the final extract is cleaner and reduces the chances of matrix effects during detection.

c)C18 (Octadecyl Silane):

Removal of Non-Polar Interferents: C18 is a non-polar sorbent material used to remove non-polar interferences, such as lipids and fatty substances, which are often present in food matrices. These non-polar compounds can interfere with the detection of analytes by HPLC or other analytical methods.

Improved Selectivity: By removing fats, oils, and other non-polar substances, C18 helps improve the selectivity and sensitivity of the analysis. It ensures that only the target analytes remain in the sample after cleanup, leading to more accurate and reliable results.

5.4 Identification and Quantification of Caffeine in Coffee by HPLC

Caffeine is a stimulant in coffee, tea, and medications that boosts alertness. It is identified by retention time and quantified using a calibration curve.

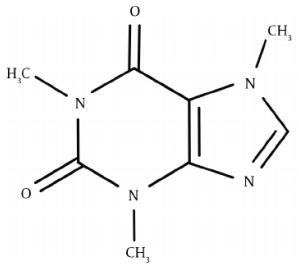


Figure 13: Structure of caffeine

5.4.1 Calibration curve for Caffeine

The Calibration curve for caffeine is given below:

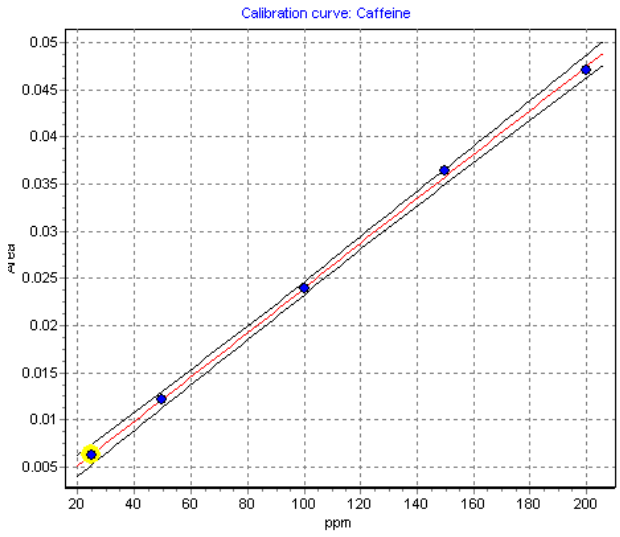


Figure 14: Calibration curve of caffeine

The calibration curve is a critical component of HPLC quantification. It establishes a linear relationship between caffeine concentration and detector response (absorbance).

- The x-axis (ppm) represents different concentrations of caffeine standards, ranging from approximately 20 ppm to 200 ppm.

- The y-axis (Absorbance at 272 nm) corresponds to the detector's response, increasing as caffeine concentration increases.
- Blue data points represent the actual measured values for each concentration.
- The red regression line shows a strong linear correlation, indicating that the HPLC system provides a consistent and proportional response to caffeine concentration.
- Two outer red lines represent the confidence interval, confirming the precision and reliability of the method.
- The yellow-highlighted data point at the lowest concentration (around 20 ppm) ensures the method's ability to detect caffeine even at low levels, signifying the limit of detection (LOD).
- The overall trend suggests an excellent linear fit, which quantifies unknown caffeine concentrations in actual samples based on their peak areas.

5.4.2 HPLC curve for Coffee

The HPLC curve for Coffee is given below

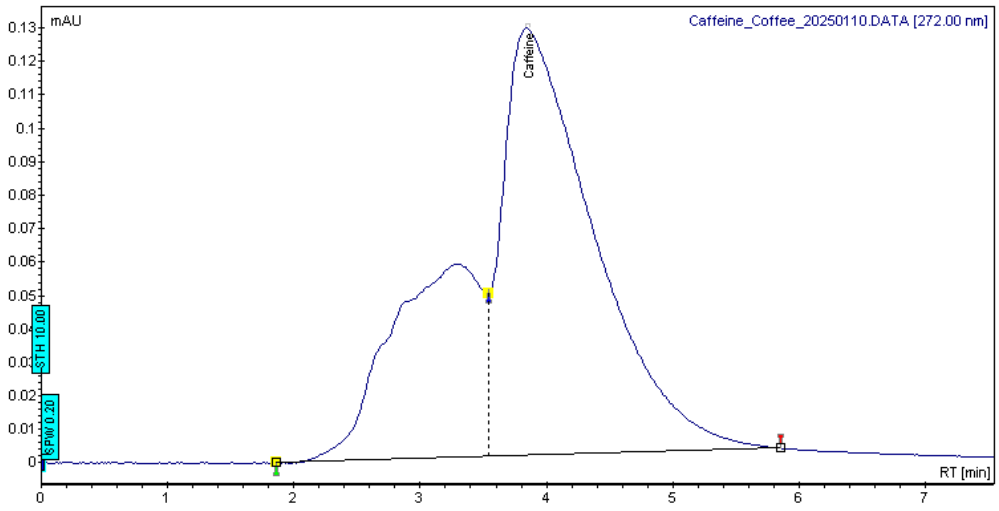


Figure 15: Chromatogram of Caffeiine in Coffee

The chromatogram visually represents the separation and detection of caffeine in a coffee sample. Each peak corresponds to a detected compound, with retention time (RT) indicating when it was eluted from the column.

- The x-axis (RT in minutes) indicates the time required for caffeine to pass through the column and reach the detector.
- The y-axis (Absorbance in mAU) represents the signal intensity at 272 nm, where caffeine absorbs UV light most efficiently.
- The sharp peak at RT ~4.2 min, labeled "Caffeine," confirms the presence of caffeine in the sample. This retention time should match that of a pure caffeine standard under the same chromatographic conditions.
- The peak shape is symmetric, suggesting good separation efficiency, minimal peak tailing, and a well-optimized mobile phase composition.
- The smaller peaks appearing before and after caffeine are likely due to other organic compounds present in the coffee, but their lower intensities indicate they are in much smaller quantities.
- The dotted vertical line at the peak apex marks the maximum absorbance, which is used to determine the peak area (AUC), a key factor for quantification.
- A red marker near RT 6 min suggests a quality control marker or an impurity that did not interfere with caffeine detection.

Observations

- The calibration curve shows strong linearity, enabling precise quantification of caffeine based on absorbance.

- The chromatogram validates caffeine identification through retention time and peak shape.

- Minimal interference from other compounds suggests a clean extraction and efficient chromatographic separation.

- The area under the caffeine peak will be compared to the calibration curve to determine the caffeine concentration in the coffee sample.

5.5 Identification and Quantification of Caffeine in soft drinks

5.5.1 HPLC curve for SOFT DRINKS

The HPLC curve for soft drinks is given blow:

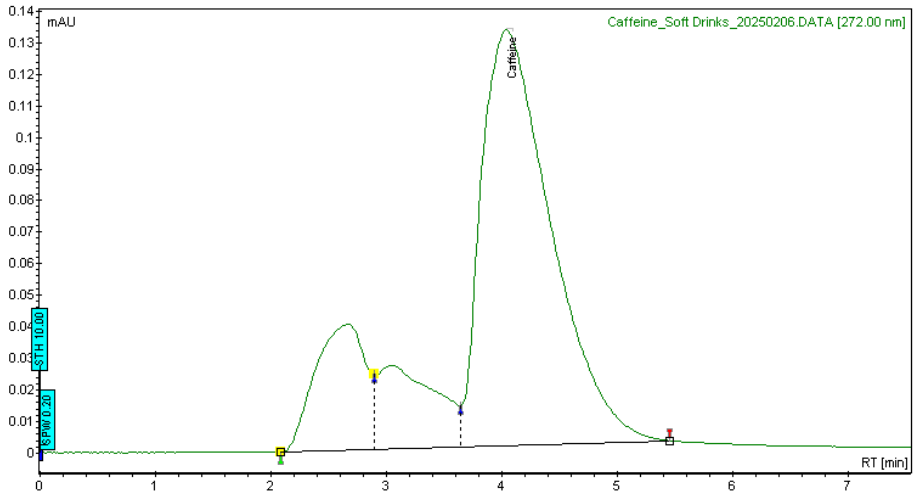


Figure 16: Chromatogram of caffeine in Soft drinks

The HPLC chromatogram illustrates the detection and quantification of caffeine in a soft drink sample. The analysis was performed at 272 nm, wavelength caffeine exhibits strong UV absorption. The separation of caffeine and other components in the soft drink matrix is based on retention time (RT), with each peak representing a compound detected by the system.

- The x-axis (RT in minutes) represents the time required for each compound to elute from the HPLC column.
- The y-axis (Absorbance in mAU) denotes the detector's response, with peak height correlating to the concentration of each compound.
- A sharp peak at RT ~4.2 minutes confirms the presence of caffeine, as its retention time matches that of the caffeine standard.
- The caffeine peak is well-defined and symmetrical, indicating efficient separation and minimal interference from other components.
- Smaller peaks between RT ~2 to 3.5 minutes suggest the presence of additional compounds in the soft drink, possibly preservatives, sugars, or flavouring agents.

- The yellow and purple markers highlight peaks of minor components, likely due to beverage additives or matrix effects.
- The dotted vertical lines mark the peak maxima, which is essential for calculating retention times and peak area (AUC).
- A red marker at RT ~6 minutes could indicate a trace impurity or another beverage component, but it does not interfere with caffeine detection.
- The stable baseline indicates minimal background noise and an optimized mobile phase, ensuring precise and accurate quantification.

Observation:

- The caffeine retention time (~4.2 min) in the soft drink corresponds with that in the coffee sample, confirming method consistency.

- Smaller peaks appearing before caffeine indicate additional compounds in the soft drink matrix, which may impact caffeine solubility or extraction efficiency.

- The area under the caffeine peak will be analyzed against the calibration curve to determine the caffeine concentration in the soft drink.

- The chromatogram showcases the effective separation of caffeine, highlighting HPLC's reliability in detecting caffeine across various sample types.

Chapter 6

Result and Discussion

The caffeine quantification in coffee and soft drink samples was performed using High-Performance Liquid Chromatography (HPLC), which revealed significant differences in caffeine concentrations across the two matrices. The results underscore the substantial variation in caffeine content, highlighting the distinctive properties of coffee and soft drinks regarding caffeine concentration.

The caffeine content was quantified in coffee at 29119 ppm (29119 mg/kg), equating to 29.119 mg of caffeine per gram of coffee powder. This finding is consistent with the established literature, which suggests that coffee, particularly Robusta varieties, contains high concentrations of caffeine [4]. When 1 g of coffee was dissolved in 100 mL of hot water to prepare a typical 100 mL cup, the caffeine content was determined to be 29.119 mg per cup, reinforcing the expected caffeine levels in coffee beverages.

Conversely, the caffeine content in the soft drink sample was found to be 104.04 ppm (104.04 mg/L). This value aligns with typical caffeine concentrations in commercially available soft drinks, where caffeine is synthetically added in regulated amounts. The markedly lower caffeine content in soft drinks, as compared to coffee, reflects the controlled nature of caffeine addition in such beverages, ensuring that they do not exceed safe consumption levels.

The caffeine extraction efficiency was optimized using a sonication-based method, which enhances the recovery of caffeine from complex matrices [9]. The method demonstrated high efficacy, as indicated by the well-resolved chromatographic peaks observed in coffee and soft drink samples. The sharpness of these peaks suggests minimal interference from other compounds in the matrix, ensuring the accuracy of the caffeine quantification.

In the chromatographic analysis, minor peaks were observed in the soft drink samples, which can be attributed to additional components such as preservatives, flavoring agents, and sweeteners. However, these compounds did not interfere with the quantification of caffeine, further validating the robustness of the HPLC method. The distinct separation of caffeine from these additives underscores the method's suitability for complex beverage matrices.

The stark contrast in caffeine concentrations between coffee and soft drinks highlights the more significant stimulant effect of coffee. A 100 mL serving of coffee contains approximately three times the caffeine found in the same soft drink volume. This substantial difference emphasizes the potential for higher caffeine intake from coffee, particularly for individuals consuming multiple servings daily. In contrast, while soft drinks are less potent sources of caffeine, their widespread consumption, particularly among adolescents, may still contribute significantly to overall caffeine intake. Excessive caffeine consumption has been associated with various health concerns, including insomnia, increased heart rate, and anxiety [4].

The precision and reliability of the HPLC method, enhanced by fluorescence detection, were evident in the clear and reproducible results obtained for both caffeine matrices. The minimal baseline noise and well-defined peaks in the chromatograms indicate that the method provides highly accurate caffeine quantification, even in complex matrices like coffee and soft drinks. These findings support the continued application of HPLC for routine analysis of caffeine in food and beverage products, ensuring both compliance with safety standards and the prevention of excessive caffeine intake.

These results contribute to the growing body of knowledge on caffeine quantification and its implications for public health. As the consumption of caffeinated beverages continues to rise globally, it is crucial to understand the variability in caffeine concentrations across different products and to assess the cumulative caffeine intake in the population. Further studies are needed to explore the impact of different brewing techniques, coffee bean origins, and beverage formulations on caffeine levels, as well as the potential long-term health consequences of excessive caffeine consumption.

Chapter 7

Conclusion

The analysis of caffeine content in coffee and soft drinks using HPLC provided critical insights into caffeine distribution in different beverage matrices, emphasizing the substantial variation in concentration. The measured caffeine content in coffee (29119 ppm or 29119 mg/kg) highlights its role as a potent caffeine source, with 29.119 mg per gram of coffee powder and 29.119 mg per 100 mL serving, reinforcing the strong stimulant effect of brewed coffee. In contrast, the caffeine concentration in soft drinks was significantly lower **(**104.04 ppm or 104.04 mg/L**),** reflecting the controlled addition of synthetic caffeine in commercial beverages. The chromatographic analysis confirmed the reliability of the extraction and quantification methods, with well-defined caffeine peaks and minimal interference from matrix components. The presence of additional peaks in soft drinks suggests the influence of additives or beverage composition on caffeine solubility. This study underscores the importance of accurate caffeine quantification for regulatory compliance and consumer safety, particularly considering excessive caffeine intake risks. The findings further validate HPLC as a robust analytical tool for caffeine determination across various food matrices. Future research should explore the impact of brewing methods, coffee bean origin, and beverage formulation on caffeine concentration, as well as potential improvements in extraction and detection methodologies to enhance analytical sensitivity and efficiency.

References

[1] IARC. (2012). Aflatoxins. International Agency for Research on Cancer. https://monographs.iarc.fr/wp-content/uploads/2018/06/mono100F-1.pdf

[2] Anastassiades, M., Lehotay, S. J., Stajnbaher, D., & Schenck, F. J. (2003). Fast and easy multiresidue method employing acetonitrile extraction/partitioning and "dispersive solid-phase extraction" for the determination of pesticide residues in fruits and vegetables. *Journal of AOAC International*, 86(2), 412-431. https://doi.org/10.1093/jaoac/86.2.412

[3] Sartori, A. V., de Mattos, J. S., de Moraes, M. H. P., & da Nóbrega, A. W. (2015). Determination of aflatoxins M1, M2, B1, B2, G1, and G2 and ochratoxin A in UHT and powdered milk by modified QuEChERS method and ultra-high-performance liquid chromatography tandem mass spectrometry. *Food Analytical Methods*, 8, 2321–2330. https://doi.org/10.1007/s12161-015-0128-4

[4] EFSA. (2015). Scientific opinion on the safety of caffeine. European Food Safety Authority. *EFSA Journal*, 13(5), 4102. https://www.efsa.europa.eu/en/efsajournal/pub/4102

[5] Russo, M., Dugo, P., Fanali, C., Dugo, L., Zoccali, M., Mondello, L., & De Gara, L. (2018). Use of an online extraction technique coupled to liquid chromatography for determination of caffeine in coffee, tea, and cocoa. *Food Analytical Methods*, 11, 2637–2644. https://doi.org/10.1007/s12161-018-1247-5

[6] Gliszczyńska-Świgło, A., & Rybicka, I. (2015). Simultaneous determination of caffeine and water-soluble vitamins in energy drinks by HPLC with photodiode array and fluorescence detection. *Food Analytical Methods*, 8, 139–146. https://doi.org/10.1007/s12161-014-9880-0

[7] Drunkler, D. A., Morais, D. N., Massarolo, K. C., Ardohain, E. N. G., Lima, J. F., & Ferreira, F. D. (2023). Method for determination of multi-mycotoxins in milk: QuEChERS extraction modified followed by HPLC-FL analysis. *Food Analytical Methods*, 17, 47–60. https://doi.org/10.1007/s12161-023-02550-0

[8] Miklós, G., Angeli, C., Ambrus, Á., Nagy, A., Kardos, V., Zentai, A., Kerekes, K., Farkas, Z., Jóźwiak, Á., & Bartók, T. (2020). Detection of aflatoxins in different matrices and food-chain positions. *Frontiers in Microbiology*, 11, 1916. https://doi.org/10.3389/fmicb.2020.01916

[9] Chirfa, G., Merdassa, Y., & Gure, A. (2020). Salting-out assisted liquid–liquid extraction for analysis of caffeine and nicotinic acid in coffee by HPLC–UV/Vis detector. *Journal of Analysis and Testing*, 4, 298–306. https://doi.org/10.1007/s41664-020-00148-7

[10] Sirhan, A. Y., Tan, G. H., Al-Shunnaq, A., Abdulra'uf, L., & Wong, R. C. S. (2013). QuEChERS-HPLC METHOD FOR AFLATOXIN DETECTION OF DOMESTIC AND IMPORTED FOOD IN JORDAN. *Journal of Liquid Chromatography & Related Technologies*, *37*(3), 321–342. https://doi.org/10.1080/10826076.2012.745138

[11] Atlabachew, M., Abebe, A., Alemneh Wubieneh, T., & Tefera Habtemariam, Y. (2021). Rapid and simultaneous determination of trigonelline, caffeine, and chlorogenic acid in green coffee bean extract. *Food science & nutrition*, *9*(9), 5028–5035. https://doi.org/10.1002/fsn3.2456

Appendix



Figure 17: Sonicating mobile phase of aflatoxin



Figure 18: Syringe filtering



Figure 19: Sonicating Caffeine sample

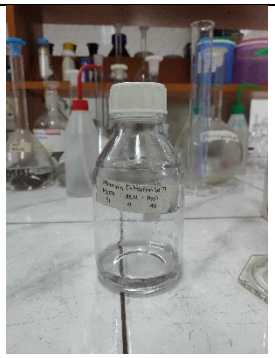


Figure 20: Aflatoxin Extraction Solvent

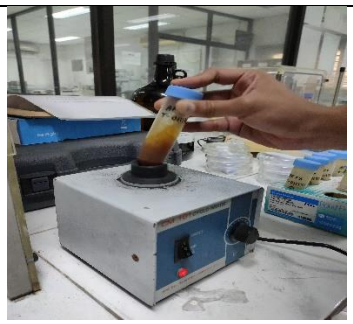


Figure 21: Vortexing Aflatoxin Sample



Figure 22: Sample collected for Aflatoxin Determination



Figure 23: Centrifuging Aflatoxin Sample



Figure 24: Micro pipetting Methanol

Table 6: Appendix